# WORLD POPULATION WEIGHTS AND THE US INTEREST RATE

A Plain-Language Guide to: *An Extended Model of Weights of Countries in the World Population and the US Interest Rate*

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This guide explains a new academic paper in plain language — no equations, no jargon. Just the key ideas, why they matter, and what policymakers should do about them.

## 1. The Puzzle

Every country on Earth watches when the United States Federal Reserve (the US central bank) changes its interest rates. When the Fed raises rates — as it did dramatically in 2022–2023 — it affects borrowing costs, currencies, and economic growth everywhere, not just in America.

But here is the puzzle: **the same Fed rate hike hits different countries very differently.** Brazil suffers more than Germany. Singapore shakes harder than China. Why?

Previous research focused on factors like trade links, currency regimes, or financial integration. This paper adds something new: **population size**.

| 558 bps | 41 bps |
| --- | --- |
| *Singapore's effective shock (0.07% of world population)* | *India's effective shock (17.8% of world population)* |

*Both countries face the same 100-basis-point Fed hike. The difference? Population weight.*

## 2. Why Does Population Size Matter?

Think of the world financial system as a swimming pool. The United States is a large country that owns much of the pool. When it makes waves (changes rates), the water moves. Now:

- **Large countries (India, China):** They are so big that they move along with the water — the wave barely displaces them relative to their own weight. They also have deep, liquid financial markets that absorb shocks.
- **Small countries (Portugal, Singapore):** They are like small boats. The same wave tosses them much more violently. They must raise their own interest rates significantly just to prevent capital fleeing to the US.

The paper shows mathematically that there is a **tipping point** at roughly 0.5% of world population. Below that threshold, countries enter a zone of amplified vulnerability — a feedback loop where smallness makes them even more sensitive to US monetary conditions.

### How the Numbers Look

| Country | World Pop. Share | Extra Rate Shock | Vulnerability |
|---|---|---|---|
| China | 17.5% | 41 bps | **LOW** |
| India | 17.8% | 41 bps | **LOW** |
| USA | 4.2% | 198 bps | **MEDIUM** |
| Indonesia | 3.5% | 231 bps | **MEDIUM** |
| Brazil | 2.7% | 280 bps | **MEDIUM-HIGH** |
| Germany | 1.1% | 389 bps | **HIGH** |
| Portugal | 0.13% | 512 bps | **VERY HIGH** |
| Singapore | 0.07% | 558 bps | **VERY HIGH** |

*bps = basis points (1% = 100 bps). Shock assumes a 100 bps Fed hike.*

## 3. The Strategic Game Between Countries

Central banks do not just react to the Fed — they also respond to each other. If India raises its rates, Brazil may feel pressure to do likewise to prevent capital outflows. This creates a strategic interaction that the paper models as a formal game.

### The Nash Approach (Each Country Acts Selfishly)

In a Nash equilibrium — the classic solution in game theory — every country sets its rate to be as good as possible for itself, taking everyone else's rate as given. The result?

- Every country sets its rate **too high**.
- The average global interest rate ends up at **5.11%** — higher than necessary.
- Everyone is worse off. This is the classic **prisoner's dilemma** of international policy.

### The Kant Approach (What if Everyone Did What I Do?)

The philosopher Immanuel Kant's categorical imperative says: act only according to rules you would want everyone to follow. In 2010, economist John Roemer translated this into game theory. The paper applies it here.

In a Kant equilibrium, each central bank asks: *"If all countries adopted my interest rate, would that be good?"* The answer leads to a globally cooperative rate of **3.91%** — significantly lower than the Nash rate of 5.11%.

| | Nash (Selfish) | Kant (Cooperative) |
|---|---|---|
| **Global Rate** | 5.11% | 3.91% |
| **Welfare (all)** | Worse | Better |
| **Gain from shift** | — | 27-34% for all |
| **Who gains most?** | — | Small economies |

## 4. What Should Policymakers Do?

| | |
|---|---|
| ● **For the US Federal Reserve** | Add a measure of world inflation to its interest rate decision-making process. A modest weight (10-25%) on global conditions would reduce harmful spillovers without compromising the Fed's domestic mandate. |
| ● **For the IMF** | Publish a 'Population-Weighted Spillover Monitor' alongside its annual Spillover Report, showing how Fed decisions hit countries at each point of the demographic distribution. |
| ● **For the G20** | Trial a Kantian Coordination Protocol: each central bank publicly discloses the interest rate it would choose if all other countries adopted the same rate. This transparency nudges the system toward the cooperative equilibrium. |
| ● **For Small Open Economies** | Countries representing less than 0.5% of world population should consider pre-emptive capital flow management tools, and seek regional monetary integration to pool demographic weight. |

## 5. The Bigger Picture

By 2050, Africa and South Asia will together account for over half of world population. India has already surpassed China as the world's most populous country. Nigeria is projected to be the third largest. The centre of demographic gravity is shifting dramatically.

Economic models that treat the world as a single uniform bloc — or worse, that treat it as an extension of OECD economies — will increasingly mis-describe global monetary dynamics. This paper argues that **demographic weight must become a first-class variable in international macroeconomics**.

The mathematics of Kantian cooperation adds an ethical dimension that is usually absent from technical economics: it shows formally that international cooperation is not just morally desirable but *economically rational* when policymakers think universally.

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